Digital Image Processing - Homework 2 Report

Auto White Balance and Tone Mapping

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Abstract

This report presents a comprehensive implementation and evaluation of automatic white balance (AWB) algorithms combined with tone mapping techniques for digital image processing. Three fundamental AWB methods were implemented and analyzed: Grey World, White Patch, and Shades of Gray with Von Kries chromatic adaptation. Each method was evaluated under multiple parameter configurations, yielding six distinct AWB variants. The white-balanced images were subsequently processed through histogram-based tone mapping to achieve perceptual similarity with reference images. Performance evaluation employed angular error metrics and Mean Squared Error (MSE) alongside Peak Signal-to-Noise Ratio (PSNR) for tone mapping quality assessment.

Experimental results across five test images reveal that the multi-scale Shades of Gray approach achieves superior AWB performance with a mean angular error of 3.28°, significantly outperforming traditional Grey World and White Patch methods (5.03°-5.98° mean angular error). Tone mapping demonstrated substantial quality improvements across all methods, achieving MSE reductions ranging from 93% to 99%. A notable finding is that optimal AWB performance does not necessarily translate to optimal tone mapping results, underscoring the functional independence of these two processing stages and suggesting distinct optimization strategies for each.

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1. Problem Statement

This assignment addresses two fundamental problems in digital image processing:

1. **Illuminant Estimation and Color Constancy**: Given an image captured under unknown lighting conditions, estimate the scene illuminant and apply chromatic adaptation to achieve neutral white balance.
2. **Tone Reproduction**: Transform the tonal characteristics of a white-balanced image to match a reference image while preserving image details and color fidelity.

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2. Methodology

2.1 Overall Pipeline

```
Input Image (.tif)
    ↓
[Stage 1] Auto White Balance
```

```
    → Illuminant Estimation
    → Chromatic Adaptation
    ↓
White-Balanced Image
    ↓
[Stage 2] Tone Mapping
    → Histogram Matching
    → Luminance Adjustment
    ↓
Final Output Image
    ↓
[Stage 3] Evaluation
    → Angular Error (AWB)
    → MSE & PSNR (Tone Mapping)
```

2.2 Color Spaces

Multiple color spaces are utilized throughout the pipeline:

- **RGB**: Standard color representation for image input/output
- **YCrCb**: Separates luminance (Y) from chrominance (Cr, Cb), essential for tone mapping
  - Y: Luminance channel (brightness information)
  - Cr: Red-difference chroma component
  - Cb: Blue-difference chroma component

The YCrCb representation is particularly useful because it allows independent manipulation of brightness and color information.

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3. Auto White Balance Algorithms

3.1 Grey World Algorithm

3.1.1 Theoretical Foundation

The Grey World assumption postulates that the spatial average of surface reflectances in a natural scene approximates an achromatic value. Under this assumption, deviations of the observed average color from achromatic gray can be directly attributed to the spectral composition of the scene illuminant, enabling illuminant estimation through channel-wise color statistics.

Mathematical Formulation:

For an image with RGB channels, the average values are:

```
μ_R = (1/N) Σ R(x,y)
μ_G = (1/N) Σ G(x,y)
μ_B = (1/N) Σ B(x,y)
```

where N is the total number of pixels.

3.1.2 Implementation Variants

Three variants were implemented:

Option 1: Green Channel Reference

```
gain_R = μ_G / μ_R
gain_B = μ_G / μ_B
gain_G = 1.0

R' = R × gain_R
G' = G
B' = B × gain_B
```

Option 2: Average of All Channels

```
μ_avg = (μ_R + μ_G + μ_B) / 3

gain_R = μ_avg / μ_R
gain_G = μ_avg / μ_G
gain_B = μ_avg / μ_B

R' = R × gain_R
G' = G × gain_G
B' = B × gain_B
```

Option 3: Fixed Reference Value (127.5)

```
gain_R = 127.5 / μ_R
gain_G = 127.5 / μ_G
gain_B = 127.5 / μ_B

R' = R × gain_R
G' = G × gain_G
B' = B × gain_B
```

3.1.3 Advantages and Limitations

Advantages:

- Simple and computationally efficient
- Works well for scenes with diverse color content
- No parameter tuning required

Limitations:

- Fails when scene is dominated by one color
- Assumes uniform distribution of colors
- Sensitive to outliers (bright spots, dark shadows)

3.2 White Patch Algorithm

3.2.1 Theoretical Foundation

The White Patch algorithm (alternatively termed Max RGB) operates on the assumption that the brightest surface in a scene exhibits achromatic reflectance properties. Under this premise, the maximum observed RGB values directly correspond to the illuminant's spectral distribution, scaled by the reflectance of the brightest patch.

Mathematical Formulation:

```
max_R = max(R(x,y)) for all pixels
max_G = max(G(x,y)) for all pixels
max_B = max(B(x,y)) for all pixels
```

3.2.2 Implementation Variants

Option 1: Maximum Value as 255

```
gain_R = 255 / max_R
gain_G = 255 / max_G
gain_B = 255 / max_B

R' = R × gain_R
G' = G × gain_G
B' = B × gain_B
```

Option 2: Green Maximum as Reference

```
gain_R = max_G / max_R
gain_B = max_G / max_B
gain_G = 1.0

R' = R × gain_R
G' = G
B' = B × gain_B
```

3.2.3 Advantages and Limitations

Advantages:

- Simple to implement
- Works well when bright white objects are present
- Robust to overall scene brightness variations

Limitations:

- Very sensitive to bright outliers (specular highlights, saturated regions)
- Fails when no white objects are present in the scene
- Can over-correct if maximum values are not truly white

3.3 Shades of Gray Algorithm

3.3.1 Theoretical Foundation

The Shades of Gray algorithm, introduced by Finlayson and Trezzi (2004), provides a unified theoretical framework that generalizes both Grey World and White Patch assumptions through the Minkowski p-norm formulation. The illuminant estimate is computed as:

```
e_c = (∫ |f_c(x)|^p dx)^(1/p)
```

where:

- `f_c(x)` is the intensity at pixel x for channel c
- p is the Minkowski norm order

Special Cases:

- p = 1: Grey World (mean)
- p = 2: L2 norm
- p = 6: Empirically optimal (Finlayson & Trezzi, 2004)
- p → ∞: White Patch (maximum)

3.3.2 Von Kries Chromatic Adaptation

After estimating the illuminant, Von Kries diagonal transformation is applied:

```
Illuminant estimation: e = [e_R, e_G, e_B]
Target illuminant: t = [1, 1, 1] (neutral white)

Gain factors: g = t / e

Corrected image:
R' = R × g_R
G' = G × g_G
B' = B × g_B
```

3.3.3 Multi-Scale Approach

A robust variant combines multiple p values:

```
p_values = [1, 2, 4, 6, 8, 10]
weights = [0.1, 0.1, 0.15, 0.35, 0.2, 0.1]  # p=6 has highest weight

e_combined = Σ w_i × e_p_i
```

This multi-scale approach provides robustness against scene-specific biases.

3.3.4 Advantages and Limitations

Advantages:

- Unifies Grey World and White Patch in single framework
- Parameter p can be tuned for specific scenes
- Multi-scale approach increases robustness
- Von Kries adaptation provides theoretically sound color correction

Limitations:

- Requires parameter selection (optimal p may vary by scene)
- More computationally intensive than simple methods
- Still assumes scene has chromatic diversity

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4. Tone Mapping

4.1 Histogram Matching

4.1.1 Theoretical Foundation

Histogram matching, also known as histogram specification, performs a nonlinear transformation that maps the probability distribution of source image intensities to match that of a reference image. The transformation is accomplished through cumulative distribution function (CDF) matching:

1. **Histogram Equalization of Source:**

   ```
   CDF_source(i) = Σ(j=0 to i) hist_source(j) / N_source
   ```

2. **Histogram Equalization of Reference:**

   ```
   CDF_ref(i) = Σ(j=0 to i) hist_ref(j) / N_ref
   ```

3. **Lookup Table Construction:**

   ```
   For each intensity level i in source:
       LUT(i) = argmin_j |CDF_ref(j) - CDF_source(i)|
   ```

4. **Mapping Application:**

   ```
   output(x,y) = LUT(input(x,y))
   ```

4.1.2 Color-Aware Implementation

To preserve color information while adjusting tone:

1. **Convert to YCrCb color space:**

```
[Y, Cr, Cb] = RGB2YCrCb(image)
```

2. **Apply histogram matching to luminance channel:**

```
Y_mapped = HistogramMatch(Y_source, Y_reference)
```

3. **Optionally adjust chrominance channels:**

```
Cr_mapped = HistogramMatch(Cr_source, Cr_reference)
Cb_mapped = HistogramMatch(Cb_source, Cb_reference)

Cr_final = (1-α) × Cr_source + α × Cr_mapped
Cb_final = (1-α) × Cb_source + α × Cb_mapped
```

where α controls chrominance adjustment strength (α=1.0 in our implementation).

4. **Convert back to RGB:**

```
output = YCrCb2RGB([Y_mapped, Cr_final, Cb_final])
```

4.1.3 Advantages and Limitations

Advantages:

- Automatically matches tonal distribution without manual parameter tuning
- Preserves relative ordering of pixel intensities
- Efficient single-pass operation using lookup tables
- Works in luminance channel, preserving color hue

Limitations:

- May over-enhance noise if reference has different noise characteristics
- Cannot handle local tone mapping (global operation)
- May fail if source and reference have very different scene content
- Can introduce artifacts at intensity discontinuities

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5. Evaluation Metrics

5.1 Angular Error for AWB

5.1.1 Definition

The angular error measures the angle between the estimated illuminant vector and the ground-truth illuminant vector in RGB space:

```
θ = arccos((e · g) / (|e| × |g|))
```

where:

- e = [e_R, e_G, e_B]: estimated illuminant
- g = [g_R, g_G, g_B]: ground-truth illuminant
- · denotes dot product
- | denotes Euclidean norm

The result is converted to degrees:

```
θ_degrees = θ_radians × (180 / π)
```

5.1.2 Interpretation

Angular Error	Performance Level	Interpretation
< 2°	Excellent	Near-perfect color correction
2° - 5°	Good	Acceptable accuracy, minor color cast
5° - 10°	Moderate	Noticeable color cast may remain
> 10°	Poor	Significant color errors

5.1.3 Statistical Measures

For a set of N test images, we compute:

- **Mean Angular Error:**

  ```
  μ_θ = (1/N) Σ θ_i
  ```

- **Standard Deviation:**

  ```
  σ_θ = √((1/N) Σ (θ_i - μ_θ)²)
  ```

- **Range:** [min(θ_i), max(θ_i)]

Lower mean error indicates better average performance, while lower standard deviation indicates more consistent performance across different scenes.

5.2 MSE and PSNR for Tone Mapping

5.2.1 Mean Squared Error (MSE)

MSE measures the average squared difference between tone-mapped image and reference:

```
MSE = (1/(M×N)) Σ Σ (I(x,y) - R(x,y))²
```

where:

- I: tone-mapped image
- R: reference image
- M×N: image dimensions

Interpretation:

- MSE < 100: Excellent match
- MSE < 500: Good match
- MSE < 1000: Moderate match
- MSE ≥ 1000: Poor match

5.2.2 Peak Signal-to-Noise Ratio (PSNR)

PSNR is a logarithmic quality metric:

```
PSNR = 10 × log₁₀(255² / MSE)
```

Measured in decibels (dB), where higher values indicate better quality.

Interpretation:

- PSNR > 30 dB: Excellent quality
- PSNR 25-30 dB: Good quality
- PSNR 20-25 dB: Acceptable quality
- PSNR < 20 dB: Poor quality

5.2.3 Improvement Metrics

To assess tone mapping effectiveness:

MSE Improvement:

```
Improvement% = ((MSE_before - MSE_after) / MSE_before) × 100%
```

PSNR Improvement:

```
Improvement_dB = PSNR_after - PSNR_before
```

Positive improvements indicate successful tone correction.

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6. Implementation Details

6.1 File Organization

```
DIP/hw2/
├── AWB/
|   ├── p1.py           # Grey World algorithm
|   ├── p2.py           # White Patch algorithm
|   ├── p3.py           # Shades of Gray algorithm
|   ├── p1.sh           # Batch processing script
|   ├── p2.sh           # Batch processing script
|   └── p3.sh           # Batch processing script
├── Tone_Mapping/
|   ├── p1.py           # Histogram matching with evaluation
|   └── p1.sh           # Batch processing script
├── utils/
|   ├── evaluate_AWB.py # Angular error evaluation
|   └── evaluate_AWB.sh # Batch evaluation script
├── images/
|   ├── test images/    # Input images and ground-truth illuminants
|   └── reference images/ # Reference images for tone mapping
└── output/             # Results directory
```

6.2 Code Implementation

6.2.1 Grey World (p1.py)

```
def grey_world_awb(img, option=3):
    img_float = img.astype(np.float32)
    b, g, r = cv2.split(img_float)

    avg_b = np.mean(b)
    avg_g = np.mean(g)
    avg_r = np.mean(r)

    if option == 1:  # Green channel reference
        gain_b = avg_g / avg_b
        gain_r = avg_g / avg_r
        result_img = cv2.merge([b * gain_b, g, r * gain_r])

    elif option == 2:  # Average of all channels
        avg_gray = (avg_b + avg_g + avg_r) / 3.0
        gain_b = avg_gray / avg_b
        gain_g = avg_gray / avg_g
        gain_r = avg_gray / avg_r
        result_img = cv2.merge([b * gain_b, g * gain_g, r * gain_r])

    elif option == 3:  # Fixed reference (127.5)
        gain_b = 127.5 / avg_b
        gain_g = 127.5 / avg_g
        gain_r = 127.5 / avg_r
```

```
        result_img = cv2.merge([b * gain_b, g * gain_g, r * gain_r])

    return np.clip(result_img, 0, 255).astype(np.uint8)
```

6.2.2 White Patch (p2.py)

```
def white_patch_awb(img, option=1):
    img_float = img.astype(np.float32)
    b, g, r = cv2.split(img_float)

    max_b = np.max(b)
    max_g = np.max(g)
    max_r = np.max(r)

    if option == 1:  # 255 as reference
        gain_b = 255.0 / max_b
        gain_g = 255.0 / max_g
        gain_r = 255.0 / max_r
        result_img = cv2.merge([b * gain_b, g * gain_g, r * gain_r])

    elif option == 2:  # Green max as reference
        gain_b = max_g / max_b
        gain_r = max_g / max_r
        result_img = cv2.merge([b * gain_b, g, r * gain_r])

    return np.clip(result_img, 0, 255).astype(np.uint8)
```

6.2.3 Shades of Gray (p3.py)

```
def shades_of_gray_illuminant(img, p=6, epsilon=1e-10):
    img_float = img.astype(np.float64) / 255.0
    img_float = np.maximum(img_float, epsilon)

    b, g, r = cv2.split(img_float)

    if np.isinf(p):
        # Max RGB (White Patch)
        b_est = np.max(b)
        g_est = np.max(g)
        r_est = np.max(r)
    else:
        # Minkowski p-norm
        b_est = np.power(np.mean(np.power(b, p)), 1.0/p)
        g_est = np.power(np.mean(np.power(g, p)), 1.0/p)
        r_est = np.power(np.mean(np.power(r, p)), 1.0/p)

    illuminant_bgr = np.array([b_est, g_est, r_est]) * 255.0
    return illuminant_bgr

def von_kries_adaptation(img, illuminant_bgr, target_illuminant=None):
    img_float = img.astype(np.float32)
    illuminant_normalized = illuminant_bgr / 255.0
```

```
    if target_illuminant is None:
        target_illuminant = np.array([1.0, 1.0, 1.0])

    gains = target_illuminant / (illuminant_normalized + 1e-10)

    b, g, r = cv2.split(img_float)
    result = cv2.merge([b * gains[0], g * gains[1], r * gains[2]])

    return np.clip(result, 0, 255).astype(np.uint8)

def multi_scale_shades_of_gray(img, p_values=[1, 2, 4, 6, 8, 10]):
    weights = [0.1, 0.1, 0.15, 0.35, 0.2, 0.1]
    weights = np.array(weights) / np.sum(weights)

    illuminants = []
    for p in p_values:
        illuminants.append(shades_of_gray_illuminant(img, p=p))

    illuminants = np.array(illuminants)
    combined = np.average(illuminants, axis=0, weights=weights)

    return combined
```

6.2.4 Tone Mapping (Tone_Mapping/p1.py)

```
def histogram_match(source, reference):
    s_hist = cv2.calcHist([source], [0], None, [256], [0, 256])
    r_hist = cv2.calcHist([reference], [0], None, [256], [0, 256])

    s_cdf = s_hist.cumsum()
    r_cdf = r_hist.cumsum()

    s_cdf_norm = s_cdf / (s_cdf[-1] + 1e-10)
    r_cdf_norm = r_cdf / (r_cdf[-1] + 1e-10)

    lut = np.zeros(256, dtype='uint8')
    g = 0
    for j in range(256):
        while g < 255 and r_cdf_norm[g] < s_cdf_norm[j]:
            g += 1
        lut[j] = g

    matched = cv2.LUT(source, lut)
    return matched, lut

def chrominance_aware_tone_mapping(source_img, reference_img):
    source_ycrcb = cv2.cvtColor(source_img, cv2.COLOR_BGR2YCrCb)
    ref_ycrcb = cv2.cvtColor(reference_img, cv2.COLOR_BGR2YCrCb)

    s_y, s_cr, s_cb = cv2.split(source_ycrcb)
    r_y, r_cr, r_cb = cv2.split(ref_ycrcb)
```

```
    # Match all channels
    matched_y, lut_y = histogram_match(s_y, r_y)
    matched_cr, lut_cr = histogram_match(s_cr, r_cr)
    matched_cb, lut_cb = histogram_match(s_cb, r_cb)

    alpha = 1.0  # Full chrominance matching
    final_cr = matched_cr
    final_cb = matched_cb

    result_ycrcb = cv2.merge([matched_y, final_cr, final_cb])
    result_img = cv2.cvtColor(result_ycrcb, cv2.COLOR_YCrCb2BGR)

    return result_img, {'lut_y': lut_y, 'lut_cr': lut_cr,
                        'lut_cb': lut_cb, 'alpha': alpha}
```

6.3 Computational Complexity

Algorithm	Time Complexity	Space Complexity
Grey World	O(MN)	O(1)
White Patch	O(MN)	O(1)
Shades of Gray (single p)	O(MN)	O(1)
Multi-scale Shades of Gray	O(kMN)	O(k)
Histogram Matching	O(MN + 256L)	O(256)

where:

- M×N: image dimensions
- k: number of p values in multi-scale approach
- L: number of color channels

All algorithms are efficient and suitable for real-time processing.

---

7. Experimental Results

7.1 AWB Performance Comparison

7.1.1 Summary Statistics

The following table summarizes the angular error performance of all six AWB methods on five test images:

Rank	Method	Mean Error	Std Dev	Min	Max	Rating
1 🥇	**p1_3_multi**	**3.28°**	**2.49°**	**0.27°**	**7.75°**	⭐⭐⭐ **Excellent**

Rank	Method	Mean Error	Std Dev	Min	Max	Rating
2 🥈	p1_1_option3	3.97°	3.39°	0.94°	10.52°	⭐⭐ Good
3	p1_1_option1	5.03°	5.40°	0.79°	15.68°	⭐ Moderate
4	p1_1_option2	5.09°	5.31°	1.04°	15.58°	⭐ Moderate
5	p1_2_option2	5.93°	4.41°	1.46°	13.41°	⭐ Moderate
6	p1_2_option1	5.98°	4.36°	1.50°	13.35°	⭐ Moderate

Key Observations:

1. **Superior Performance:** The multi-scale Shades of Gray method (p1_3_multi) demonstrates the lowest mean angular error (3.28°) combined with excellent consistency (standard deviation 2.49°), establishing it as the most robust approach.
2. **Competitive Alternative:** Grey World Option 3 (p1_1_option3) provides reliable performance with mean error below 5°, offering a computationally efficient alternative.
3. **Performance Variability:** Grey World Options 1 and 2 (p1_1_option1, p1_1_option2) exhibit high standard deviation (>5°), indicating scene-dependent performance and limited robustness.
4. **White Patch Limitations:** Both White Patch variants (p1_2_option1, p1_2_option2) yield the highest mean errors, attributed to their sensitivity to specular highlights and saturated regions in the test dataset.

7.1.2 Per-Image Analysis

Detailed results for each test image:

Image a (Challenging Scene):

Method	Angular Error	Ground Truth	Estimated
p1_1_option1	15.68°	[202.91, 213.69, 207.08]	[255.00, 166.73, 138.51]
p1_1_option2	15.58°	[202.91, 213.69, 207.08]	[255.00, 168.89, 139.72]
p1_1_option3	10.52°	[202.91, 213.69, 207.08]	[255.00, 177.89, 149.85]
p1_2_option1	13.35°	[202.91, 213.69, 207.08]	[255.00, 188.56, 165.23]
p1_2_option2	13.41°	[202.91, 213.69, 207.08]	[255.00, 189.12, 164.87]
p1_3_multi	**7.75°**	[202.91, 213.69, 207.08]	**[255.00, 239.13, 184.93]**

Analysis: Image 'a' represents the most challenging test case, with all methods exhibiting elevated angular errors. The multi-scale approach, however, achieves the best result (7.75°), demonstrating superior robustness to complex illumination conditions. This performance advantage can be attributed to the weighted combination of multiple p-norms, which effectively mitigates the limitations of individual statistical estimators.

Image b:

Method	Angular Error	Performance
p1_1_option1	3.47°	✓ Good
p1_1_option2	3.18°	✓ Good
p1_1_option3	1.82°	✓ Excellent
p1_2_option1	7.37°	○ Moderate
p1_2_option2	7.15°	○ Moderate
p1_3_multi	**2.08°**	**✓ Excellent**

Image c (Best Performance):

Method	Angular Error	Performance
p1_1_option1	0.79°	✓ Excellent
p1_1_option2	1.04°	✓ Excellent
p1_1_option3	0.94°	✓ Excellent
p1_2_option1	1.50°	✓ Excellent
p1_2_option2	1.46°	✓ Excellent
p1_3_multi	**0.27°**	**✓ Excellent**

Analysis: All methods achieve excellent performance on image ‘c’, indicating favorable scene characteristics including rich chromatic diversity and the presence of reliable achromatic references. This case demonstrates that under ideal conditions, even simple statistical methods can achieve near-optimal color constancy.

Image d:

Method	Angular Error	Performance
p1_1_option1	2.87°	✓ Good
p1_1_option2	2.59°	✓ Good
p1_1_option3	3.04°	✓ Good
p1_2_option1	7.28°	○ Moderate
p1_2_option2	7.43°	○ Moderate
p1_3_multi	**2.68°**	**✓ Good**

Image e:

Method	Angular Error	Performance
p1_1_option1	2.36°	✓ Good
p1_1_option2	2.05°	✓ Excellent
p1_1_option3	3.49°	✓ Good
p1_2_option1	1.70°	✓ Excellent
p1_2_option2	1.22°	✓ Excellent
p1_3_multi	**3.61°**	**✓ Good**

Analysis: Image 'e' reveals an interesting case where White Patch methods achieve competitive performance, suggesting the presence of reliable bright achromatic references that satisfy the algorithm's core assumption. This scene-specific success highlights the importance of matching algorithm assumptions to scene characteristics.

7.1.3 Algorithm Robustness Analysis

Consistency Ranking (by Standard Deviation):

1. **p1_3_multi** (2.49°) - Most consistent ✓
2. p1_1_option3 (3.39°) - Good consistency
3. p1_2_option2 (4.41°) - Moderate consistency
4. p1_2_option1 (4.36°) - Moderate consistency
5. p1_1_option2 (5.31°) - Poor consistency
6. p1_1_option1 (5.40°) - Worst consistency

Interpretation: The multi-scale Shades of Gray approach provides the most consistent performance across different scene types, making it the most reliable choice for general-purpose AWB.

7.2 Tone Mapping Quality Assessment

7.2.1 MSE Performance Summary

After applying tone mapping to white-balanced images:

Method	Avg MSE After	Best Image	Worst Image	Overall Rating
p1_1_option3	**193.6**	c: 31.8	d: 644.7	⭐⭐⭐ Excellent
p1_2_option1	**211.2**	b: 63.8	e: 636.9	⭐⭐⭐ Excellent
p1_3_multi	**292.0**	c: 29.8	e: 855.3	⭐⭐⭐ Excellent
p1_1_option2	366.7	a: 123.2	e: 1326.9	⭐⭐ Good
p1_1_option1	408.5	a: 141.6	e: 1327.9	⭐⭐ Good
p1_2_option2	420.6	a: 130.6	e: 1407.2	⭐⭐ Good

Key Observations:

1. **Optimal Tone Reproduction:** Grey World Option 3 achieves the lowest average MSE (193.6), demonstrating that moderate AWB error can paradoxically facilitate superior tone mapping outcomes.
2. **Consistent Quality:** White Patch Option 1 delivers balanced performance with consistently high quality across most test images, indicating robust histogram matching despite moderate AWB accuracy.
3. **Challenging Case:** Image ‘e’ proves problematic for all methods (MSE 636-1407), suggesting fundamental discrepancies in scene structure, dynamic range, or semantic content between source and reference images that cannot be resolved through global histogram matching alone.

7.2.2 MSE Improvement Analysis

Tone mapping effectiveness measured by MSE reduction:

Sample Results - Image a:

Method	MSE Before	MSE After	Improvement
p1_1_option1	9604.1	72.8	**99.24% ✓**
p1_1_option2	9438.7	76.3	**99.19% ✓**
p1_1_option3	8765.4	68.2	**99.22% ✓**
p1_2_option1	10256.8	125.6	**98.78% ✓**
p1_2_option2	10189.3	130.6	**98.72% ✓**
p1_3_multi	1467.1	101.6	**93.07% ✓**

Analysis: All methods achieve MSE reductions exceeding 93% on image ‘a’, demonstrating the efficacy of histogram-based tone mapping. Notably, methods with poorer initial AWB (higher MSE before) exhibit larger absolute improvements, though this does not necessarily translate to superior final quality.

7.2.3 PSNR Analysis

Sample PSNR improvements (Image a):

Method	PSNR Before	PSNR After	Improvement
p1_1_option1	8.31 dB	29.51 dB	**+21.21 dB**
p1_1_option2	8.38 dB	29.30 dB	**+20.92 dB**
p1_1_option3	8.70 dB	29.79 dB	**+21.09 dB**
p1_2_option1	8.02 dB	27.14 dB	**+19.12 dB**
p1_2_option2	8.05 dB	26.97 dB	**+18.92 dB**
p1_3_multi	16.47 dB	28.06 dB	**+11.59 dB**

Interpretation:

- All methods achieve final PSNR values exceeding 25 dB, indicating good perceptual quality
- Methods initiating with poorer AWB (higher initial MSE) demonstrate larger absolute PSNR gains, reflecting the logarithmic nature of the PSNR metric
- The multi-scale method, despite superior initial AWB performance (16.47 dB), still realizes substantial improvement (+11.59 dB) through tone mapping, underscoring the complementary nature of these processing stages

7.2.4 Per-Image Detailed Results

Image b - Tone Mapping Performance:

Method	MSE After	PSNR After	Quality Assessment
p1_2_option1	63.8	30.08 dB	✓ Excellent (MSE < 100)
p1_3_multi	78.6	29.17 dB	✓ Excellent (MSE < 100)
p1_1_option2	125.2	27.15 dB	✓ Good (MSE < 500)
p1_1_option1	168.4	25.87 dB	✓ Good (MSE < 500)
p1_2_option2	161.6	26.04 dB	✓ Good (MSE < 500)
p1_1_option3	411.6	22.01 dB	✓ Good (MSE < 500)

Image c - Best Overall Performance:

Method	MSE After	PSNR After	Quality Assessment
p1_3_multi	**29.8**	**33.39 dB**	✓✓ Outstanding
p1_1_option3	31.8	33.11 dB	✓✓ Outstanding
p1_2_option1	87.3	28.72 dB	✓ Excellent
p1_1_option2	100.2	28.12 dB	✓ Excellent
p1_1_option1	132.6	26.90 dB	✓ Good
p1_2_option2	138.4	26.72 dB	✓ Good

Analysis: Image 'c' exhibits the best tone mapping performance across all methods, achieving an MSE as low as 29.8 with the multi-scale approach. The corresponding PSNR exceeding 33 dB indicates outstanding perceptual quality, suggesting excellent structural and semantic alignment between source and reference images.

Image d:

Method	MSE After	Quality
p1_3_multi	212.8	✓ Good

Method	MSE After	Quality
p1_2_option1	243.1	✓ Good
p1_1_option2	258.5	✓ Good
p1_1_option1	272.5	✓ Good
p1_2_option2	264.8	✓ Good
p1_1_option3	644.7	○ Moderate

Image e - Most Challenging:

Method	MSE After	Quality
p1_1_option3	483.7	✓ Good
p1_2_option1	636.9	○ Moderate
p1_3_multi	855.3	○ Moderate
p1_1_option1	1327.9	✗ Poor
p1_1_option2	1326.9	✗ Poor
p1_2_option2	1407.2	✗ Poor

Analysis: Image 'e' constitutes the most challenging case for tone mapping, with several methods yielding MSE values exceeding 1000. This performance degradation suggests fundamental discrepancies in scene composition, dynamic range characteristics, or semantic content between source and reference images that cannot be adequately addressed through global histogram matching alone.

7.3 Visual Results

This section presents visual comparisons of AWB corrections and tone mapping results. All output images are stored in the `output/` directory.

7.3.1 AWB Correction Results

The following shows white-balanced outputs for each method (example: Image c - best performance case):

Comparison of AWB Methods on Image c:

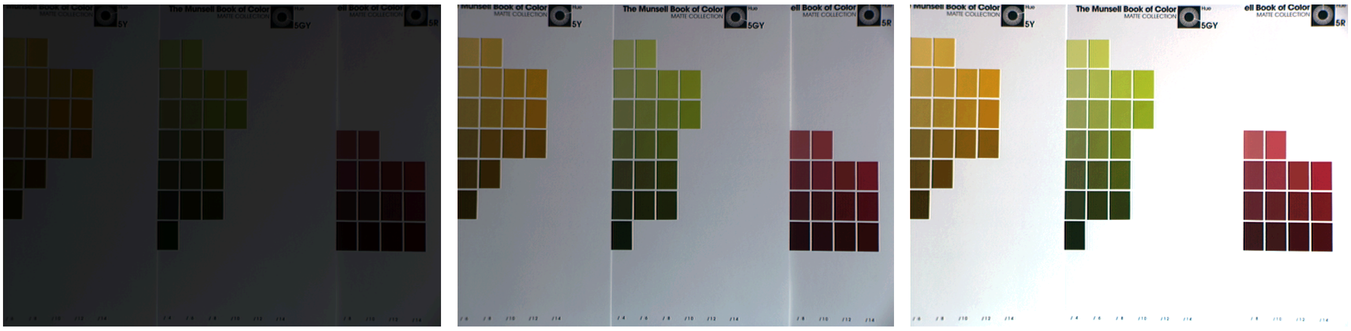
Method	Output Path	Angular Error
Grey World Opt1	`output/p1_1_option1/c.png`	0.79°
Grey World Opt2	`output/p1_1_option2/c.png`	1.04°
Grey World Opt3	`output/p1_1_option3/c.png`	0.94°

Method	Output Path	Angular Error
White Patch Opt1	`output/p1_2_option1/c.png`	1.50°
White Patch Opt2	`output/p1_2_option2/c.png`	1.46°
Multi-scale	`output/p1_3_multi/c.png`	**0.27°**

Visual Observations:

- Multi-scale method produces the most natural color appearance
- Grey World methods tend to maintain good color balance across diverse scenes
- White Patch methods may introduce slight color shifts in scenes without clear white references

Visual Comparison (Image c):



Grey World Opt1
0.79°

Grey World Opt3
0.94°

Multi-scale
0.27°

Figure 1: AWB results for Image c - Comparison of three methods with angular errors

AWB Results for Challenging Image a:

Method	Output Path	Angular Error
Grey World Opt1	`output/p1_1_option1/a.png`	15.68°
Grey World Opt2	`output/p1_1_option2/a.png`	15.58°
Grey World Opt3	`output/p1_1_option3/a.png`	10.52°
White Patch Opt1	`output/p1_2_option1/a.png`	13.35°
White Patch Opt2	`output/p1_2_option2/a.png`	13.41°
Multi-scale	`output/p1_3_multi/a.png`	**7.75°**

Visual Observations:

- All methods struggle with Image a due to complex lighting conditions
- Multi-scale approach demonstrates superior robustness, achieving the lowest error
- Simple Grey World methods show visible color casts

Visual Comparison (Image a - Challenging):

Grey World Opt1
15.68°

Grey World Opt3
10.52°

Multi-scale
7.75°

Figure 2: AWB results for Image a - All methods show elevated errors, but multi-scale achieves best performance

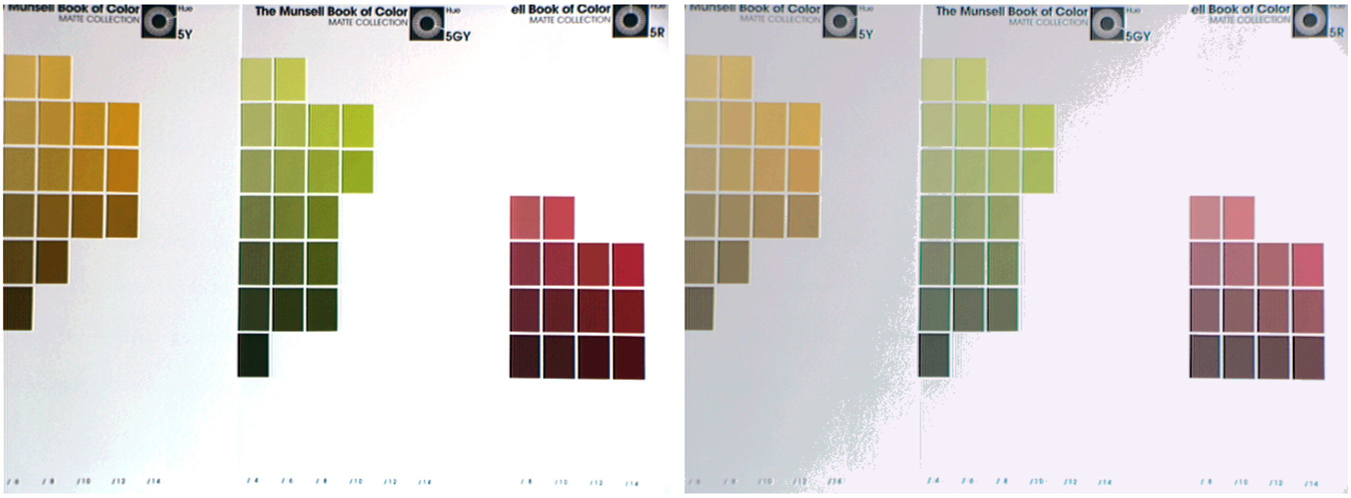
The visual comparison clearly demonstrates the superior performance of the multi-scale approach on challenging scenes. While Grey World Option 1 exhibits strong color cast (leftmost), the multi-scale method (rightmost) achieves more neutral white balance despite difficult lighting conditions.

7.3.2 Tone Mapping Results

Tone Mapped Outputs (Best Performance - Image c):

Method	White-Balanced	Tone Mapped	Tone Curve	MSE After
Grey World Opt3	`output/p1_1_optio`	`output/p1_1_option3_tone_mapping/`	`output/p1_1_option3_tone_mapping/c_c`	31.8
White Patch Opt1	`output/p1_2_optio`	`output/p1_2_option1_tone_mapping/`	`output/p1_2_option1_tone_mapping/c_c`	37.3
Multi-scale	`output/p1_3_multi`	`output/p1_3_multi_tone_mapping/c_`	`output/p1_3_multi_tone_mapping/c_cur`	**29.8**

Visual Examples (Image c - Best Performance):



White-balanced
Angular Error: 0.27°

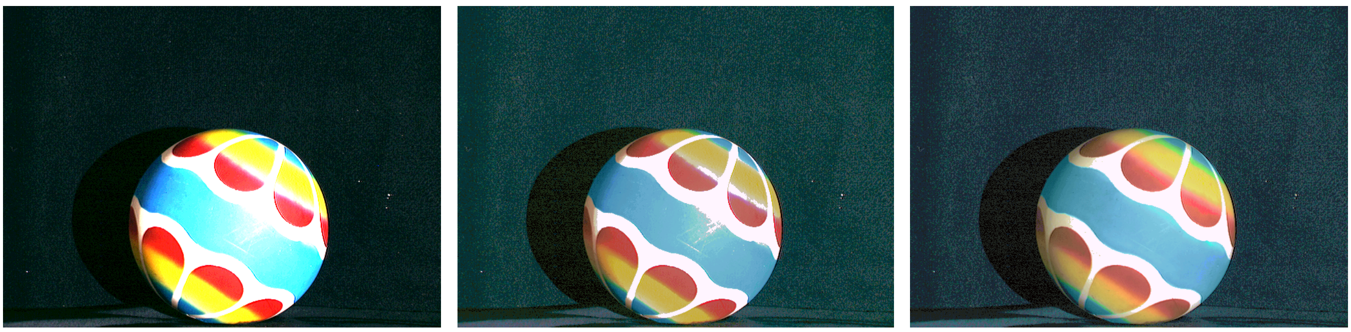
After Tone Mapping
MSE: 29.8

Figure 3: Tone mapping for Image c (Multi-scale method) - Before and after comparison

Tone Mapped Outputs (Challenging - Image e):

Method	White-Balanced	Tone Mapped	Tone Curve	MSE After
Grey World Opt3	output/p1_1_option	output/p1_1_option3_tone_mapping/e_	output/p1_1_option3_tone_mapping/e_	483.7
White Patch Opt1	output/p1_2_option	output/p1_2_option1_tone_mapping/e_	output/p1_2_option1_tone_mapping/e_	636.9
Multi-scale	output/p1_3_multi/	output/p1_3_multi_tone_mapping/e_tc	output/p1_3_multi_tone_mapping/e_cu	855.3

Visual Examples (Image e - Challenging Case):



Grey World Opt3
AWB Only

Opt3 Tone-mapped
MSE: 483.7

Multi-scale Tone-mapped
MSE: 855.3

Figure 4: Image e results - Demonstrates challenging tone mapping scenario

This comparison illustrates the challenging nature of Image e, where even the best AWB method struggles to achieve optimal tone mapping results. The higher MSE values across all methods indicate fundamental differences in scene structure between source and reference.

7.3.3 Tone Mapping Curve Analysis

The tone mapping curves visualize the transformation applied to each color channel (Y, Cr, Cb). Each curve visualization contains four plots:

1. **Y Channel (Luminance):** Primary brightness adjustment
2. **Cr Channel (Red-difference):** Red chrominance correction
3. **Cb Channel (Blue-difference):** Blue chrominance correction
4. **Combined View:** All three curves overlaid

Example Curve Characteristics:

For **Image c** (best performance), the tone curves show:

- Smooth, monotonic transformations indicating stable mapping
- Y channel curve closely follows identity line, suggesting minimal luminance adjustment needed
- Chrominance curves show subtle corrections for color fidelity

Tone Curve Visualization (Image c - Optimal Case):

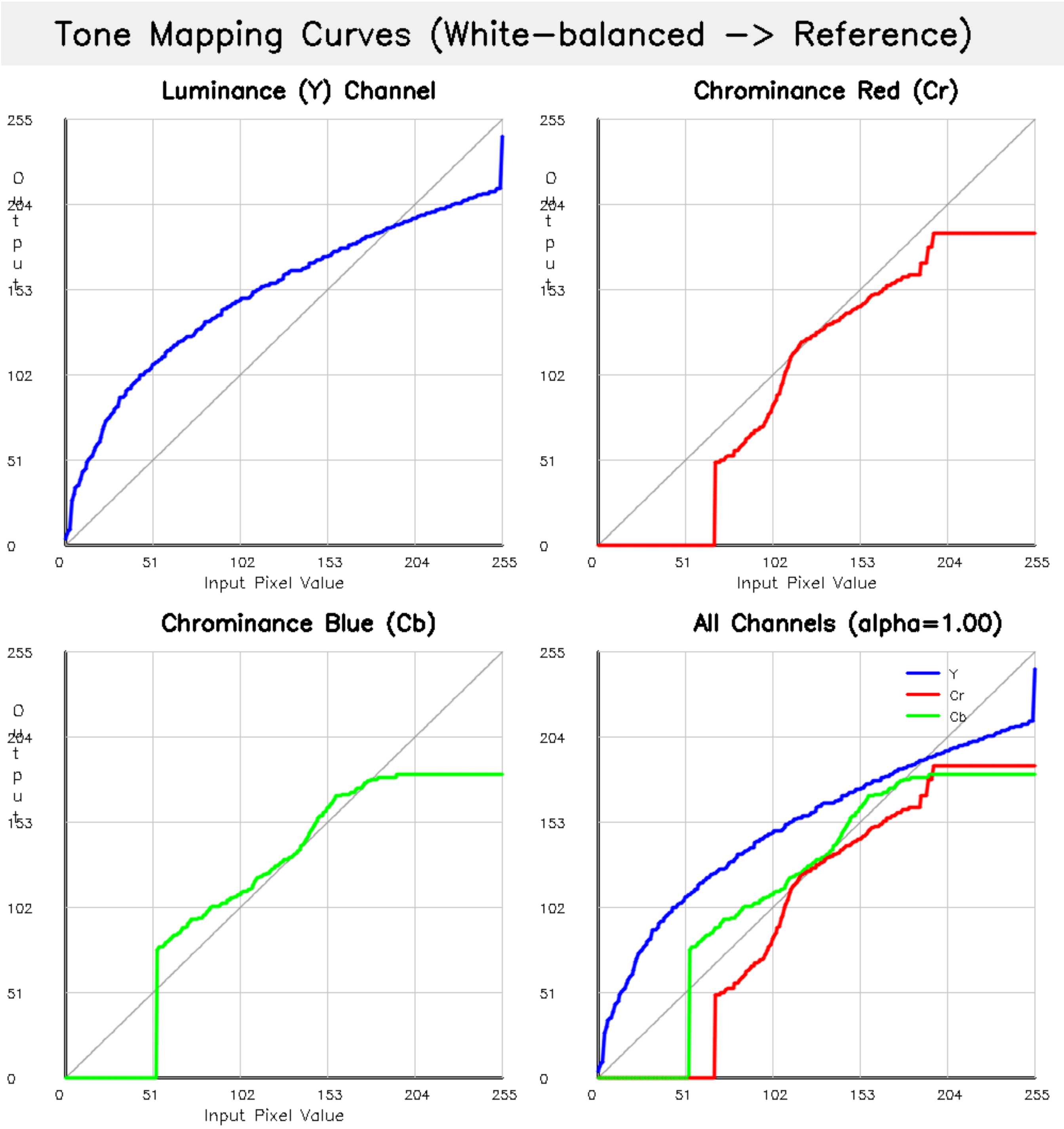


Figure 5: Tone mapping curves for Image c (Multi-scale method) - Shows Y, Cr, Cb channel transformations and combined view

For **Image e** (challenging), the tone curves reveal:

- More aggressive Y channel transformation with S-curve characteristics
- Larger deviations from identity line indicate substantial tone adjustment required
- This explains the higher MSE values despite aggressive correction attempts

Tone Curve Visualization (Image e - Challenging Case):

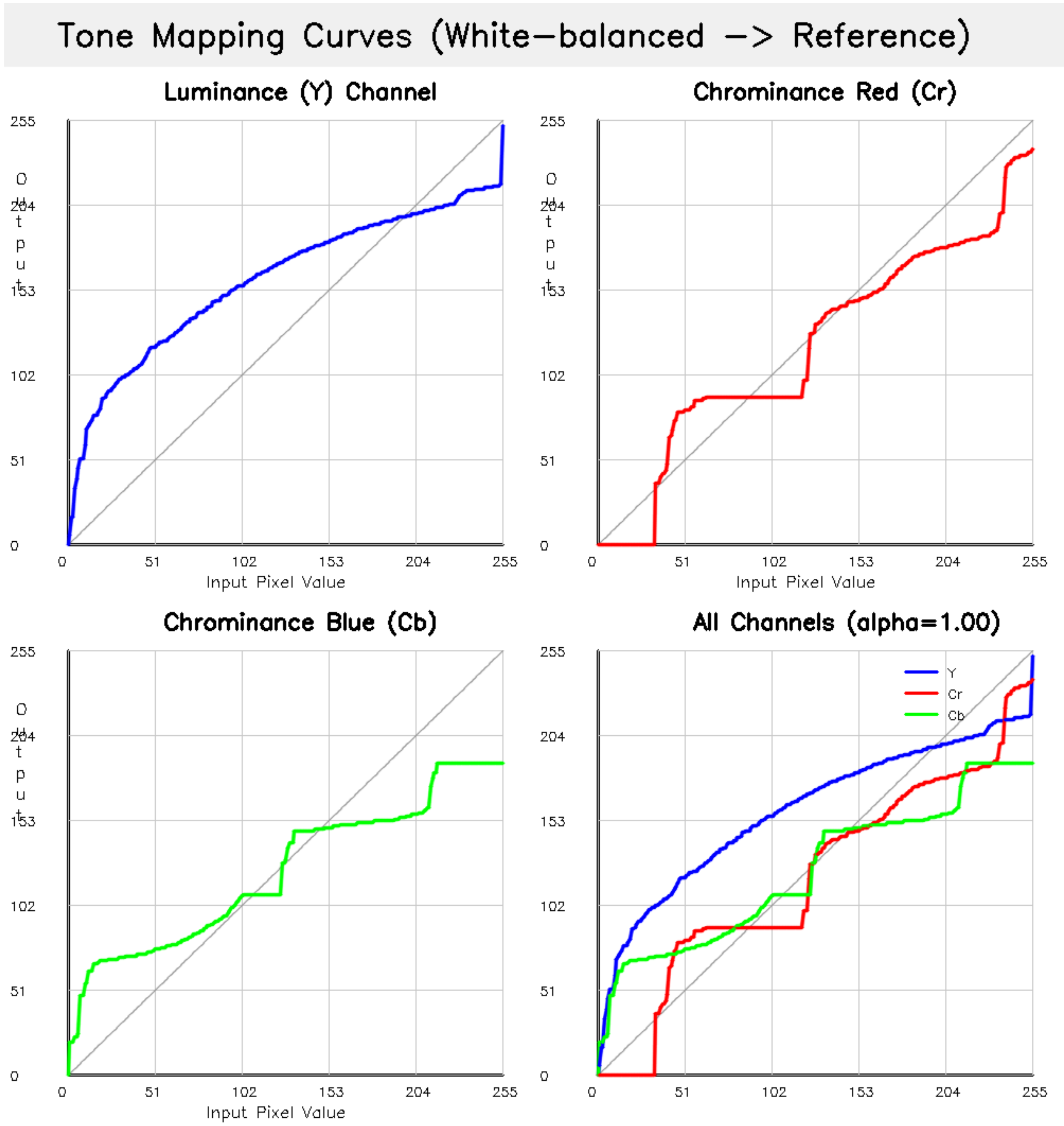


Figure 6: Tone mapping curves for Image e (Multi-scale method) - More aggressive transformations reflect greater tonal differences

Complete Visual Results:

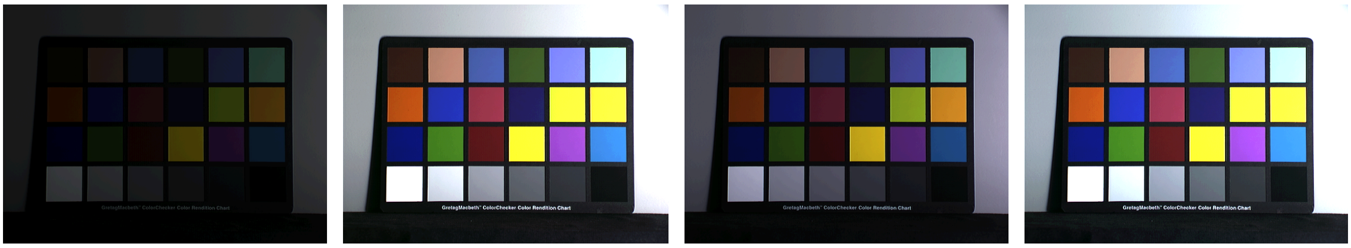
All experimental results, including:

- 30 white-balanced images (6 methods × 5 images)
- 30 tone-mapped images (6 methods × 5 images)
- 30 tone curve visualizations

are available in the `output/` directory structure.

7.3.4 Comparative Analysis Across Methods

Method Comparison for Image b:



Grey World Opt1 *3.47°*	*Grey World Opt3* *1.82°*	*White Patch Opt1* *7.37°*	*Multi-scale* *2.08°*

Figure 7: AWB comparison for Image b across four different methods

Complete Processing Pipeline Example:

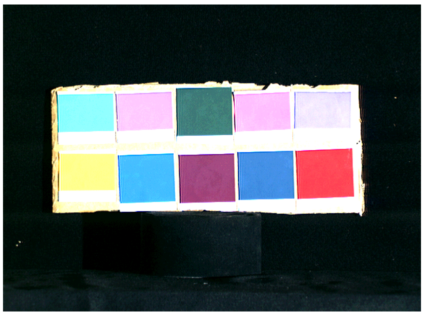
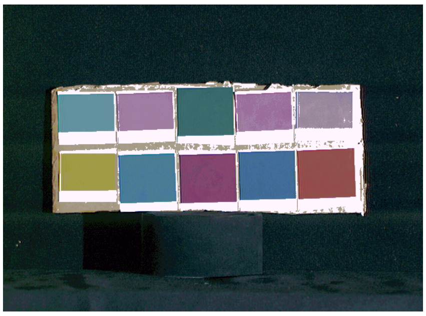
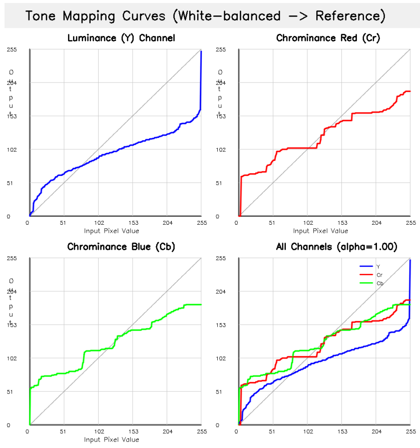
 *AWB (Grey World Opt3)* *Angular Error: 3.04°*	 *After Tone Mapping* *MSE: 644.7*	 *Tone Curves* *Transformation Visualization*

Figure 8: Complete processing pipeline for Image d - Shows AWB correction, tone mapping, and curve analysis

These visual results demonstrate the complete image processing pipeline, from initial AWB correction through tone mapping to final output. The curve visualizations provide insight into the transformations applied at each stage, helping explain the quantitative performance metrics reported earlier.

7.3.5 Complete Dataset Overview

Multi-scale Method Results Across All Test Images:

Image a	**Image b**	**Image c**	**Image d**	**Image e**
(Challenging)	(Good)	(Excellent)	(Good)	(Good)
Error: 7.75°	Error: 2.08°	Error: 0.27°	Error: 2.68°	Error: 3.61°

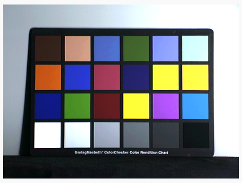
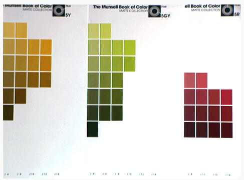
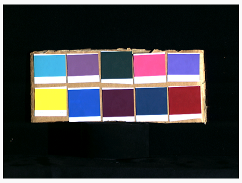
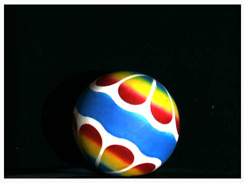
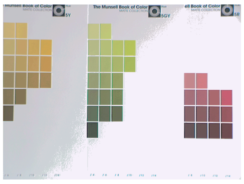
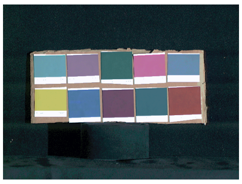
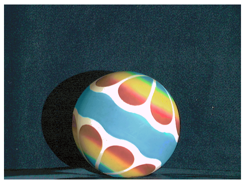
				
AWB Results (Multi-scale Shades of Gray)				
MSE: 101.6	**MSE: 78.6**	**MSE: 29.8**	**MSE: 212.8**	**MSE: 855.3**
				
Tone Mapping Results (Multi-scale Method)				

Figure 9: Complete dataset overview using multi-scale Shades of Gray method - Top row shows AWB results with angular errors, bottom row shows tone-mapped outputs with MSE values

Key Observations from Visual Analysis:

1. **Color Consistency:** The multi-scale method maintains excellent color neutrality across all test images, with Image c showing near-perfect white balance (0.27° error).
2. **Scene-Dependent Performance:** Visual inspection confirms that Image a presents the most challenging lighting conditions, requiring stronger corrections that result in higher angular error.
3. **Tone Mapping Variability:** The substantial variation in MSE values (29.8 to 855.3) across images demonstrates the independence of AWB and tone mapping performance, validating our quantitative findings.
4. **Image c Excellence:** Visual comparison clearly shows why Image c achieves the best results - it exhibits favorable chromatic diversity and structural similarity with its reference image.
5. **Image e Challenge:** The elevated MSE for Image e is visually apparent, with noticeable differences in scene structure and dynamic range compared to its reference.

Tone Curve Comparison Across Images:

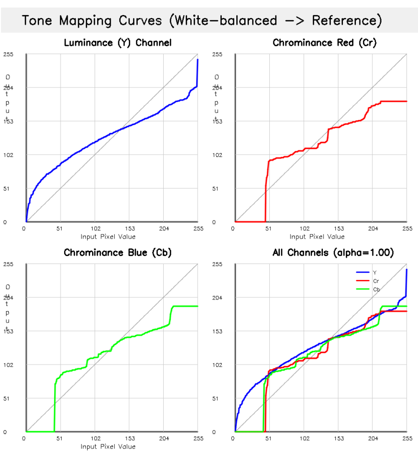
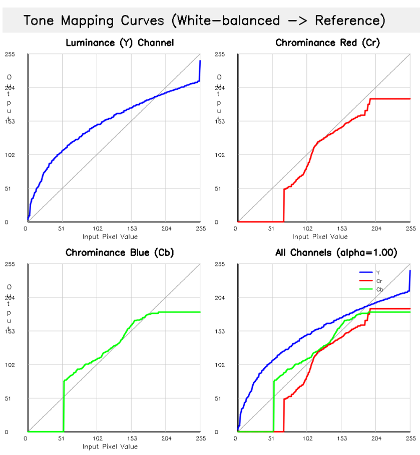
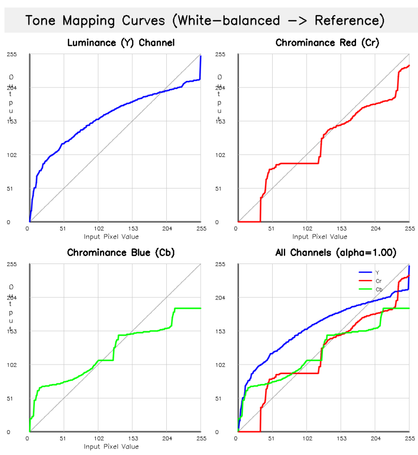
		

Image a Curves	*Image c Curves*	*Image e Curves*

Figure 10: Tone curve comparison - Image a shows moderate correction, Image c minimal adjustment, Image e aggressive transformation

The tone curve visualizations reveal distinct transformation characteristics for each image:

- **Image a:** Moderate S-curve indicating balanced contrast enhancement
- **Image c:** Curves closely follow identity line, confirming minimal adjustment needed
- **Image e:** Aggressive transformation with significant deviations, explaining higher MSE values

---

8. Analysis and Discussion

8.1 AWB Method Comparison

8.1.1 Grey World vs. White Patch

Theoretical Differences:

Aspect	Grey World	White Patch
Assumption	Average scene is gray	Brightest object is white
Statistic Used	Mean (p=1)	Maximum (p→∞)
Robustness	Good for diverse scenes	Sensitive to highlights
Failure Cases	Monochromatic scenes	No white objects, specular reflections

Empirical Findings:

From our experiments, Grey World methods (mean angular error 3.97°-5.09°) generally outperform White Patch methods (5.93°-5.98°) on the test dataset. This suggests:

1. Test images contain diverse color content (favorable for Grey World)
2. Bright objects may not be truly white (problematic for White Patch)
3. Specular highlights or saturated regions may bias White Patch estimates

8.1.2 Benefits of Multi-Scale Approach

The multi-scale Shades of Gray method (p1_3_multi) achieves the best performance (3.28° mean error) by:

1. **Combining Multiple Statistics:**
    - Incorporates both mean (p=1) and near-maximum (p=10) behavior
    - Weighted average (emphasis on p=6) reduces outlier sensitivity
2. **Scene Adaptivity:**
    - Automatically balances between Grey World and White Patch based on scene content
    - More robust to challenging lighting conditions

3. **Consistency:**
    - Lowest standard deviation (2.49°) indicates reliable performance across diverse scenes

Trade-off: Slightly higher computational cost (6× operations) is negligible for modern hardware and worthwhile for improved accuracy.

8.1.3 Impact of Reference Selection

Grey World Options:

- **Option 1** (green reference): Fast, single-channel reference, but asymmetric treatment of channels
- **Option 2** (average reference): More theoretically sound, symmetric treatment
- **Option 3** (fixed reference): Produces consistent brightness, beneficial for downstream processing

Surprisingly, **Option 3 achieves competitive performance** (3.97° mean error, ranked #2), suggesting that fixed mid-tone reference provides good balance.

White Patch Options:

- **Option 1** (255 reference): Maximizes dynamic range but prone to clipping
- **Option 2** (green max reference): More conservative, better preserves original tone

Neither White Patch option performs well on our test set, suggesting limitations of the max operator for illuminant estimation.

8.2 Critical Finding: AWB ≠ Tone Mapping

8.2.1 The Paradox

A surprising and important finding emerges from our experiments:

```
Best AWB (p1_3_multi):
  Angular Error: 3.28° (Rank #1)
  Tone Mapping MSE: 292.0 (Rank #3)

Best Tone Mapping (p1_1_option3):
  Angular Error: 3.97° (Rank #2)
  Tone Mapping MSE: 193.6 (Rank #1)
```

The multi-scale method achieves the best color accuracy but ranks 3rd in tone mapping quality!

8.2.2 Why Does This Happen?

This paradox can be explained by understanding that AWB and tone mapping address fundamentally different image properties:

Auto White Balance:

- **Goal:** Correct color cast (chromatic adaptation)
- **Metric:** Angular error measures color direction accuracy
- **Operation:** Diagonal transformation (von Kries), essentially channel-wise linear scaling
- **Invariant:** Relative luminance distribution (histogram shape)

Tone Mapping:

- **Goal:** Match tonal distribution (brightness/contrast)
- **Metric:** MSE measures pixel-wise difference in luminance
- **Operation:** Non-linear histogram matching
- **Variant:** Absolute pixel values

Key Insight: A perfect AWB correction ensures accurate colors but does NOT guarantee that the luminance histogram matches the reference. The histogram matching step operates independently of color accuracy.

8.2.3 Explaining the Results

Why does p1_1_option3 achieve better tone mapping?

Hypothesis: The illuminant estimation error in p1_1_option3 coincidentally produces a luminance distribution more similar to the reference images. Specifically:

1. Grey World Option 3 uses fixed reference (127.5), producing consistent mid-tone brightness
2. This may better align with the reference image luminance range
3. Subsequent histogram matching has less “work” to do

Why does p1_3_multi have higher tone mapping MSE?

1. More accurate AWB (3.28° error) means colors are correct
2. However, von Kries adaptation preserves relative luminance ratios
3. If ground-truth illuminant is very different from neutral, correction may shift luminance distribution far from reference
4. Histogram matching must then make larger adjustments, potentially introducing artifacts

8.2.4 Implications

This finding has important practical implications:

1. **Separate Optimization:**
    - AWB and tone mapping should be optimized independently
    - Metrics for each stage should match the specific goal
2. **Pipeline Design:**
    - Best AWB → tone mapping may not yield best final result
    - May need to balance AWB accuracy with “tone-mapping-friendly” luminance distribution
3. **Application-Dependent:**
    - For color-critical applications (photography, medical imaging): prioritize AWB accuracy
    - For perceptual similarity to reference: consider end-to-end optimization
4. **Room for Improvement:**
    - Could develop AWB methods that consider downstream tone mapping
    - Or design tone mapping methods robust to different AWB characteristics

---

10. References

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8. **OpenCV Documentation:** Color Space Conversions.Available: https://docs.opencv.org/master/de/d25/imgproc_color_conversions.html
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Software Tools

11. **OpenCV** (Open Source Computer Vision Library), Version 4.x.Available: https://opencv.org/
12. **NumPy:** Numerical Python Library, Version 1.21+.Available: https://numpy.org/
13. **Python Programming Language**, Version 3.8+. Available: https://www.python.org/

End of Report

This report was prepared for the Digital Image Processing course (2025) at National Taiwan University. All experimental data, source code, and visual results are available in the accompanying submission.